Finding links between language and psychological states

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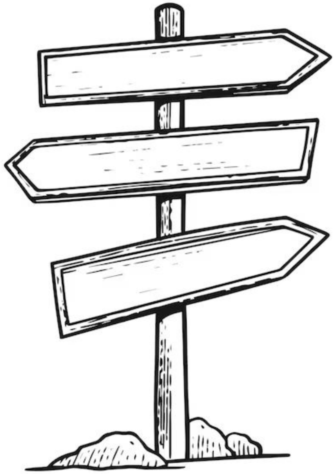
Signposting :)

1. **Tools - LIWC**
   - What it is for
   - Some shortcomings
   - How it works
2. **Resources - DAIC-WOZ**
3. **Their intersection - my master's thesis**
   - RQ's and hypotheses
   - Results
   - Implications



Part 1: Linguistic Inquiry and Word Count - LIWC

"[o]ur goal was to create a program that simply looked for and counted words in psychology-relevant categories across multiple text files" (Tausczik & Pennebaker, 2010, pp. 27)

LIWC

But why?

Language encodes and decodes human thought

"[Words] are the medium by which cognitive, personality, clinical, and social psychologists attempt to understand human beings" (Tausczik & Pennebaker, 2009, pp. 25)

The thoughts and feelings people have

…are expressed by…

The things people say

…give insight into…

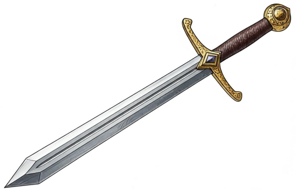
So…

possibly?

Similar mental state → Similar languaging

Only part of the puzzle:

- Nonverbal Communication:
    - Facial expressions
    - Eye contact, gaze
    - Gestures
    - Posture
- Vocal aspects:
    - Tone
    - Pitch
    - Speech rate
- Context



How does it work?

Dictionaries.

- analyses input texts word by word, checks whether each word is present within the user-selected dictionaries.
- the final output is a **percentage**, reflecting how many words within a text belong to a given dictionary category.

Category	*Examples*	*Words in category*
Total pronouns		116
Personal pronouns	I, them, her	70
1st person singular	I, me, mine	12
1st person plural	We, us, our	12
2nd person	You, your, thou	20
3rd person singular	She, her, him	17
3rd person plural	They, their, they'd	10
Impersonal pronouns	It, its, those	46
Articles	A, an, the	3
Verbs		314
Past tense	Went, ran	145
Present tense	Hear, take	169
Cognitive processes		730
Insight	Think, know	195
Causation	Because, effect	108
Discrepancy	Should, would	76
Tentative	Maybe, perhaps	155
Certainty	Always, never	83
Inhibition	Block, constrain	111
Inclusive	And, with, include	18
Exclusive	But, without	17

“Empirical results using LIWC demonstrate its ability to detect meaning in a wide variety of experimental settings, including to show attentional focus, emotionality, social relationships, thinking styles, and individual differences.”

(Tausczik & Pennebaker, 2010)

In Practice

What you need:

- A LIWC subscription (30 days = 72zł, approx. 17 EUR)
- A corpus of texts which you want to analyse (docx, csv, xlsx)
- Selected LIWC dictionaries (inbuilt LIWC English dictionary/custom)

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
participan	Text	WC	Analytic	Clout	Authentic	Tone	WPS	BigWords	Dic	Linguistic	function	pronoun	ppron	i	we
302	i'm fine ho	614	24.72	20.05	96.34	85.38	6.33	13.52	95.28	75.24	53.58	15.64	9.12	6.68	0.65
303	okay how	1968	7	79.38	92.14	86.58	19.11	9.86	97	82.57	62.3	24.85	17.73	7.72	0.15
304	i'm doing g	987	3.61	10.7	94.32	53.17	9.49	13.98	97.06	81.97	61.7	22.59	15.1	9.83	0.71
305	i'm doing a	3328	6.48	12.01	94.66	40.5	11.56	9.71	96.39	83.65	61.93	23.56	16.35	10.76	0.33
307	<laughter>	2596	10.9	5.98	93.84	49.05	13.96	11.98	95.96	80.35	60.86	18.91	11.67	8.51	1.16
310	yes it's oka	1137	6.46	4.03	92.89	43.86	7.68	12.14	96.75	80.3	58.14	20.49	13.46	9.5	0.26
312	yes. fine h	1204	2.79	9.87	93.17	74.64	9.56	13.21	96.51	80.56	63.62	22.43	13.87	8.97	0
313	sure. uh. o	751	22.26	6.25	86.65	72.38	6.59	12.65	96.94	78.7	56.32	18.24	11.19	8.26	0.4
315	alright. yes	1370	13.16	15.36	82.57	41.8	10.15	10.36	96.5	81.53	58.1	20.88	14.45	10.07	0.88
316	yes. i'm fin	633	27	2.79	87.6	84.06	4.65	12.32	94.15	78.52	57.19	17.38	12.64	10.11	0
317	yes. uh i'm	642	27.88	5.11	77.45	54.08	5.73	13.24	95.33	79.13	50.16	13.55	9.66	8.72	0.31
318	yes. i'm alr	698	14.5	7.84	91.79	88.38	10.26	12.32	93.55	77.65	59.6	18.05	10.74	8.31	0.57
319	sure. mm c	681	5.56	24.38	94.08	51.94	6.36	10.72	96.04	81.5	59.18	23.35	16.59	10.87	0.29
320	[syncing]. '	758	4.72	1	97.15	39.44	4.77	12.01	96.31	80.87	60.95	23.48	15.7	13.32	0.4
321	yeah. i'm c	918	11.41	4.06	89.93	41.08	7.52	11.98	97.06	81.15	59.37	21.02	15.9	12.53	0.33
322	yes. fine th	1568	13.79	42.31	81.63	53.87	7.88	10.14	95.54	81.7	55.23	20.73	15.31	8.99	0.45
324	yeah. yes.	871	19.75	25.8	98.97	74.03	5.31	11.37	94.49	78.3	55.91	18.71	13.55	9.07	0.34
325	yes. um i'n	1799	10.78	5.6	93.47	58.53	19.14	12.4	94.33	79.99	59.92	20.84	14.45	11.34	0.5
326	yes. i'm fin	511	20.98	6.46	92.42	99	5.16	16.05	94.13	76.13	48.53	17.42	12.72	8.41	0.2
327	yes. fine th	784	10.6	23.22	91	70.36	10.45	12.37	96.05	79.08	56.51	17.6	12.76	8.04	0.26
328	yes. <laugh	2028	5.83	51.82	78.86	81.62	14.8	10.9	96.01	79.98	60.55	22.44	14.79	7.2	1.53
330	<laughter>	482	35.17	5.55	85.98	11.72	4.23	15.77	97.72	79.46	49.79	15.98	10.58	7.88	0.21
331	yes. okay.	1477	6.62	6.11	95.64	33.47	12.21	10.09	95.94	81.11	62.15	19.23	14.01	10.83	0.34
333	yes. very w	1350	16.78	5.98	95.26	90.77	8.77	13.48	94.15	79.7	56.89	19.33	12.96	10.3	0.3
335	yes. i'm ok	1393	12.84	3.73	93.87	37.81	8.24	13.21	96.63	79.47	59.44	21.18	13.85	11.49	0.43

Part 2: Discourse Analysis Interview Corpus (Wizard of Oz) - DAIC-WOZ

https://dcapswoz.ict.usc.edu/

Gratch et al., 2014; DeVault et al., 2014

- Depression, anxiety, mood disorders

Contents

193 recorded and annotated semi-structured interviews:

- interview audio,
- transcripts,
- physiological data (Biopac - GSR, ECG, respiration),
- depth sensor positional data (Microsoft Kinect),
- participants' answers to administered questionnaires (partially)

Participants

- Greater Los Angeles metropolitan area
- 2 populations: veterans of the U.S. armed forces/general public
- N = 193 (WOZ Condition)

Questionnaires

- The Positive and Negative Affect Scale (PANAS); (Watson and Clark, 1994)
- PTSD Checklist – Civilian Version; (Blanchard et al., 1996)
- **Patient Health Questionnaire (PHQ), depression module; (Kroenke and Spitzer, 2002)**
- State-Trait Anxiety Inventory; (Spielberger et al., 1983)
- a measure of social desirability; (Li and Bagger, 2007)
- Rapport scale; (Kang and Gratch, 2012)

Wizard of Oz Condition

Wizard-of-Oz

Ellie Who's someone that's been a positive influence in your life?
Participant Uh my father.
Ellie Can you tell me about that?
Participant Yeah, he is a uh
Participant He's a very he's a man of few words
Participant And uh he's very calm
Participant Slow to anger
Participant And um very warm very loving man
Participant Responsible
Participant And uh he's a gentleman has a great sense of style and he's a great cook.
Ellie Uh huh
Ellie What are you most proud of in your life?

Part 3: His Depression, Her Depression

RQ1: *Do symptoms of depression manifest themselves in patient language in the form of linguistic markers?*

RQ2: *Do differences in the epidemiology, phenomenology, and pathophysiology of depression between men and women translate to differences in linguistic marker presence?*

Selected Linguistic Markers

- Emotional Language (Positive tone, Negative tone, Anger, Anxiety, Sadness)

- Pronouns; (Savekar et al., 2022)

- Absolutist Words; (Al-Mosaiwi and Johnstone, 2018)

Hypotheses

Depressed participants will use…

H1a: **more negatively valenced language**

H1b: **more First Person Singular Pronouns**

H1c: **more Absolutist words**

…when compared to non-depressed participants.

H2: Depressed men and women **will differ between themselves** in terms of used linguistic markers of depression. **(Non-directional)**

- Silencing the Self Theory (Jack, 2010)
- Differences in disease onset, presentation, and cause

Methods

(PHQ-8 scores)

RQ1:

N total = **142**

Depressed N = 42

Non-depressed N = 100

RQ2:

Non-dep Women N = 39

Depressed Women N = 24

Non-dep Men N = 61

Depressed Men N = 18

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
participant	Text	WC	Analytic	Clout	Authentic	Tone	WPS	BigWords	Dic	Linguistic	function	pronoun	ppron	i	we
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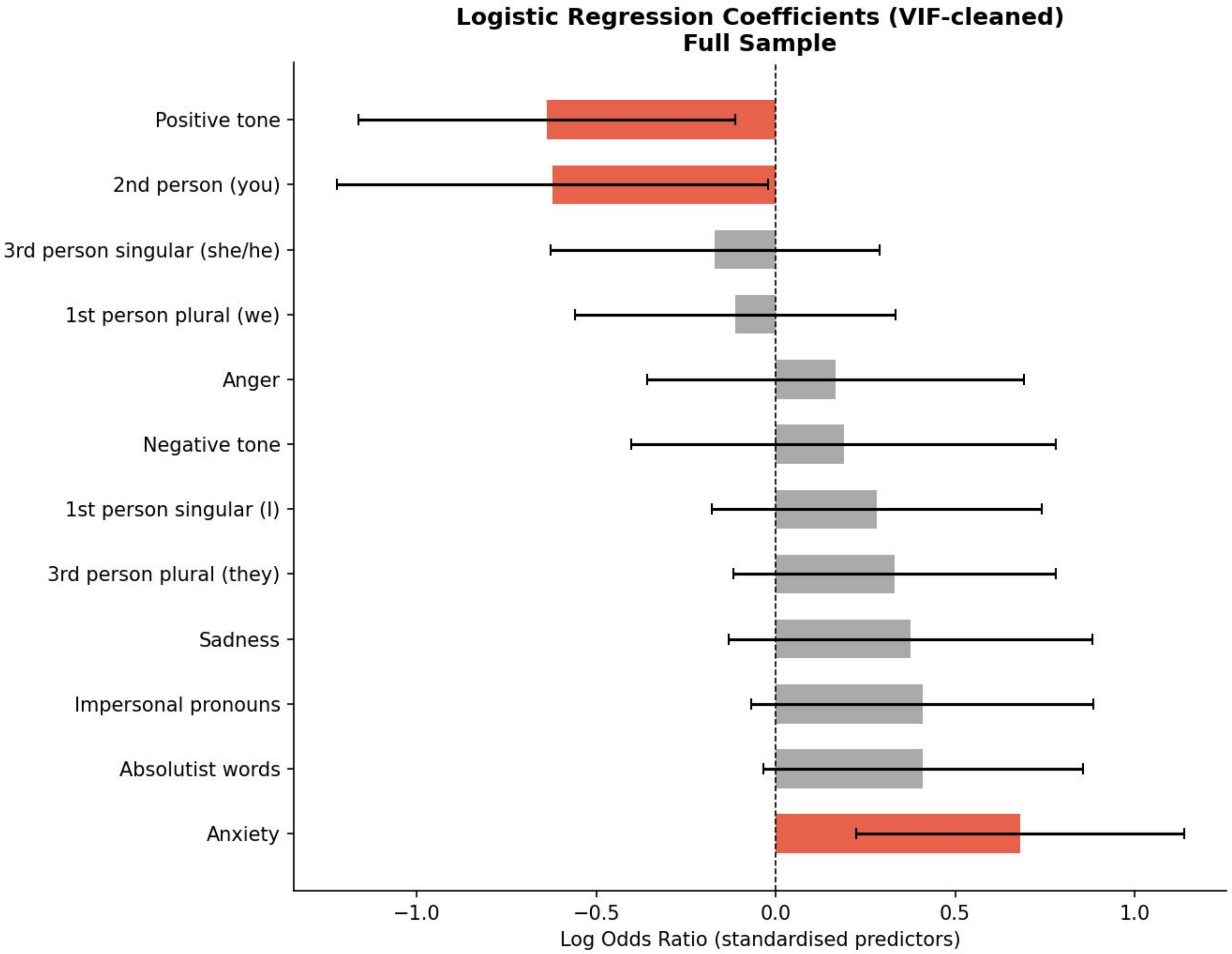
Results: RQ1 (depressed VS non-depressed)

Significant markers after T-testing:

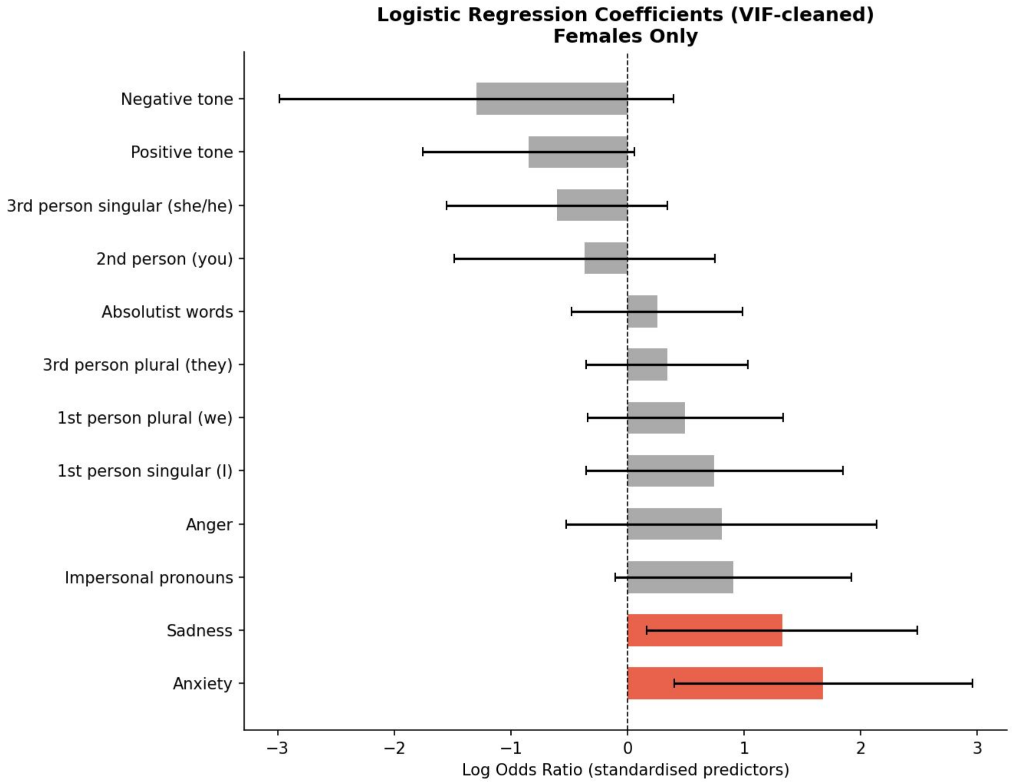
- **Anxiety** (M1 = 0.122, SD1 = 0.149 vs. M2 = 0.238, SD2 = 0.199; p raw = <0.0001, ***p* FDR = 0.003, *d* = 0.658**),
- **Negative tone** (M1 = 1.767, SD1 = 0.735 vs. M2 = 2.254, SD2 = 0.879; p raw = 0.001, ***p* FDR = 0.007, *d* = 0.602**),
- **Negative emotion** (M1 = 1.161, SD1 = 0.567 vs. M2 = 1.522, SD2 = 0.808; p raw = 0.008, ***p* FDR = 0.042, *d* = 0.518**)

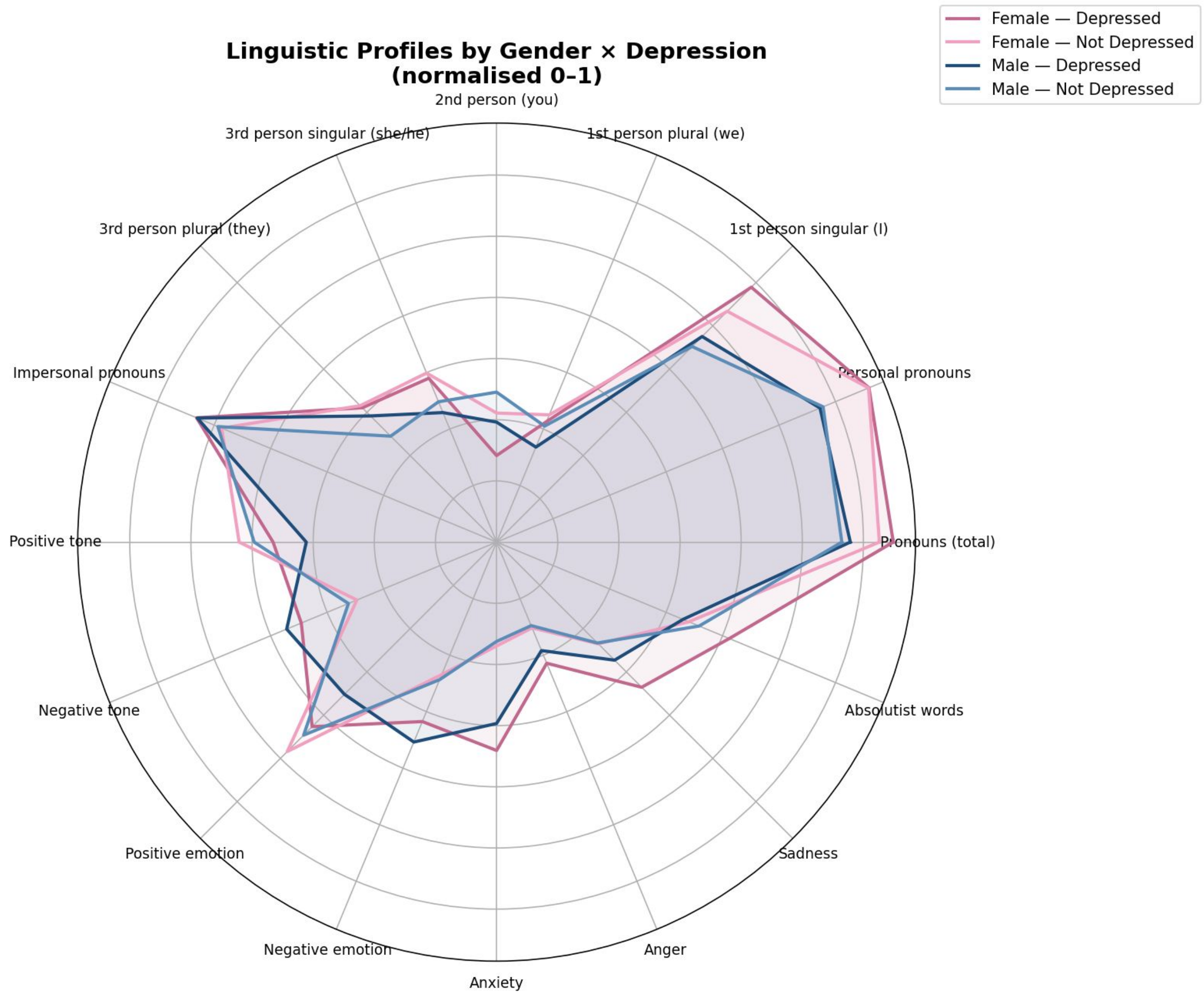


Logistic Regression Coefficients (VIF-cleaned)
Full Sample
Positive tone
2nd person (you)
3rd person singular (she/he)
1st person plural (we)
Anger
Negative tone
1st person singular (I)
3rd person plural (they)
Sadness
Impersonal pronouns
Absolutist words
Anxiety
−1.0
−0.5
0.0
0.5
1.0
Log Odds Ratio (standardised predictors)

Results: RQ2 (depressed men VS depressed women)

- Men VS Women T-testing found **no significant differences** between markers
- Further logistic regression analyses suggested that:
    - Depressed women exhibited **more Sadness and Anxiety** markers than non-depressed women
    - Depressed men and non-depressed men **did not differ significantly**





Linguistic Profiles by Gender × Depression
(normalised 0–1)
Female — Depressed
Female — Not Depressed
Male — Depressed
Male — Not Depressed
2nd person (you)
1st person plural (we)
1st person singular (I)
Personal pronouns
Pronouns (total)
Absolutist words
Sadness
Anger
Anxiety
Negative emotion
Positive emotion
Negative tone
Positive tone
Impersonal pronouns
3rd person plural (they)
3rd person singular (she/he)

Implications

H1a: Depressed participants will use more negatively valenced language when compared to non-depressed participants.

H1b: Depressed participants will use more First Person Singular Pronouns

H1c: Depressed participants will use more Absolutist words

H2: Depressed men and women will differ between themselves in terms of used linguistic markers of depression. (Non-directional)

Real-world applications of research, resources, and tools

- Bettering the understanding of mental disorders and their symptoms
- Facilitating the diagnosis of mental disorders
- Aiding the creation of mental health chat-bots, avatars

“automatic assessment of verbal and nonverbal behaviors correlated with psychological distress” (DeVault et al., 2014, pp. 254)

Thanks for your attention!

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